Ayush Asthana

Current Position:	Assistant Professor (Fall 2023–present)
Department, University:	Department of Chemistry, University of North Dakota
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GitHub:	github.com/asthanaa, github.com/asthanalab
Group website:	theasthanagroup.com
Research Areas:	Quantum chemistry, quantum computing, relativistic quantum chemistry, sustainable energy

Education

Aug 2016-Jul 2021 **PhD in Theoretical Chemistry, Johns Hopkins University**
Advisor: Prof. Lan Cheng
Dissertation: Development of relativistic quantum chemistry methods for heavy-element-containing molecules

Jul 2011- May 2016 **BS-MS Dual Degree in Chemistry, Indian Institute of Technology Kanpur**
Advisors: Prof. Debashis Mukherjee (IACS), co-advisor: Prof. Srihari Keshavamurthy (IITK)
Thesis: Multireference perturbation theory for bond breaking and quasi-degeneracy.

Employment

Aug 2023–present **Assistant Professor of Chemistry (Tenure Track)** and **Adjunct Professor of Physics**
University of North Dakota

2025–2026 (6 months) **Visiting Scientist at Pacific Northwest National Laboratory**
Host: Dr. Bo Peng
Funding: NSF EPSCoR Fellow

Aug 2021-Jul 2023 **Postdoctoral Research Associate, Virginia Tech**
Advisor: Prof. Nicholas J. Mayhall

Grants and Funding

Funding associated with my research program: $6,148,686 in project awards ($2,544,577 allocated to the Asthana Lab), plus $34,000 in in-kind computing resources.

2026–2031 **DOE Early Career Research Program (2026 award)**
Heavy Element Chemistry program
Title: *Toward Resolving the f-Electron Challenge in Heavy-Element Chemistry with Spinor Krylov Algorithms on Classical and Early Fault-Tolerant Quantum Computers*
Role: PI; Amount: **$875,000**; five years beginning Nov. 1, 2026.

2027–2029 **ACS Petroleum Research Fund, Doctoral New Investigator**
Title: *Relativistic Effects in Platinum-Catalyzed Hydrocarbon Activation*
Role: PI; Amount: **$110,000**; two years beginning Fall 2027.

2026–2029 **ARPA-E (DOE) QC3**, "Quantum-Classical Ab Initio Co-Simulation of Unconventional Superconductors"
Role: co-PI; Amount: $3,949,109 (Asthana Lab share: **$600,000**).
- Providing quantum chemistry and quantum algorithm solutions to bridge materials simulation (NREL) and hardware implementation (Infleqtion).

2026–2028 **NSF-NQVL**, "QSTD: Design: Attosecond Synchronized Photonic Entanglement Network (ASPEN-Net)"
Role: Key personnel; Asthana Lab share: **$60,000**.
- Contributing to the development of a minor in quantum information at UND.

2024–2026 **NSF EPSCoR Research Fellow**, "Quantum Algorithms for Early Fault-tolerant Quantum Computers"
Role: PI; Amount: $299,977 (Asthana Lab share: **$299,977**).
- Developing new early fault-tolerant algorithms and chemistry use cases of quantum computing.

2024–2027 **NSF ExpandQISE (Award 2427046)**, "Quantum Algorithms for Relativistic Quantum Chemistry"
Role: PI; Amount: $800,000 (Asthana Lab share: **$560,000**).
- Developing algorithms for use of quantum computing in ground and excited states for relativistic quantum chemistry.

2024–2025 **Moderna Inc.**, "Accurate and scalable molecular simulations on quantum computers"
Role: PI; Amount: $24,600 (Asthana Lab share: **$24,600**; 6 months).

2023–2025 **Early Career Scholars Program**, UND internal funding
Role: PI; Amount: $30,000 (Asthana Lab share: **$15,000**).

2023–2026 **Amazon Braket, IBM Quantum, and NSF ACCESS compute credits**
Role: PI; In-kind award: $5,000+$5,000+$12,000+$12,000 in cloud computing credits.

Publications

1. P. F. Kwao, B. Sambasivam, J. Saroni, S. P. Sundar, E. Barnes, and **A. Asthana***, "Post-Optimization Subspace Corrections to Quantum Ansätze", arXiv:2610.19109 (2026; in review).
2. Z. W. Windom*, P. F. Kwao, L. Bertels, D. Claudino*, **A. Asthana***, and R. J. Bartlett, "Resolving molecular biexcited states using classical leading-order triples corrections to quantum equation-of-motion UCCSD", arXiv:2610.09260, submitted to J. Chem. Phys. (2026).
3. J. Saroni, B. Sambasivam, **A. Asthana**, and E. Barnes, "Quantum simulation of molecular excited-state manifolds and energies using the TEPID-ADAPT-VQE algorithm", arXiv:2606.29547 (2026; in review).
4. P. F. Kwao, S. P. Sundar, B. Gupt, and **A. Asthana***, "On the generalized eigenvalue problem in subspace-based excited state methods for quantum computers", J. Chem. Theory Comput. **22**(6), 2892–2903 (2026).
5. S. P. Sundar, P. F. Kwao, A. Galda, and **A. Asthana***, "Molecular excited states using quantum subspace methods: Accuracy, resource reduction, and error-mitigated hardware implementation of q-sc-EOM", arXiv:2604.05380 (2026; in review).
6. S. P. Sundar, V. Abraham, B. Peng, **A. Asthana***, "Chemically decisive benchmarks on the path to quantum utility", arXiv:2601.10813 (2026; in review).
7. **A. Asthana***, "Exact and Tunable Quantum Krylov Subspaces via Unitary Decomposition", arXiv:2512.11788 (2026; in review).
8. H. Zhang and **A. Asthana**, "Cyclic variational quantum eigensolver: Escaping barren plateaus through staircase descent", arXiv:2509.13096 (2025; in review).
9. I. Mahmud, **A. Asthana**, M. Hoffmann and A. Abdelhadi, "Physics-Informed Neural Networks for Quantum Wavefunctions", IEEE/ICAA **1** (2024) (Conference Paper)
10. C. Zhang, X. Zheng, J. Liu, **A. Asthana**, and L. Cheng, "Analytic gradients for relativistic exact-two-component equation-of-motion coupled-cluster singles and doubles method", J. Chem. Phys. **159**, 244113 (2023).
11. A. Kumar, **A. Asthana**, V. Abraham, T. Crawford, N. Mayhall, Y. Zhang, L. Cincio, S. Tretiak, P. Dub, "Quantum Simulation of Molecular Response Properties in the NISQ Era", J. Chem. Theory Comput. **19**, 24, 9136 (2023).

Before tenure track period

12. **A. Asthana**, C. Liu, O. R. Meitei, S. E. Economou, E. Barnes, N. J. Mayhall, "Leakage Reduces Device Coherence Demands for Pulse-Level Molecular Simulations", Phys. Rev. Applied **19**, 064071 (2023).
13. **A. Asthana***, A. Kumar, V. Abraham, H. Grimsley, Y. Zhang, L. Cincio, S. Tretiak, P. Dub, S. Economou, E. Barnes, N. Mayhall, "Quantum self-consistent equation-of-motion method for computing molecular excitation energies, ionization potentials, and electron affinities", Chem. Sci., **14**, 2405 (2023).
14. A. Kumar, **A. Asthana**, C. Masteran, E. F. Valeev, Y. Zhang, L. Cincio, S. Tretiak, P. A. Dub, "Quantum Simulation of Molecular Electronic States with a Transcorrelated Hamiltonian: Higher Accuracy with Fewer Qubits", J. Chem. Theory Comput. **18**, 9, 5312 (2022).
15. J. Liu, X. Zheng, **A. Asthana**, C. Zhang, and L. Cheng, "Analytic evaluation of energy first derivatives for spin-orbit coupled-cluster singles and doubles augmented with noniterative triples method: General formulation and an implementation for first-order properties", J. Chem. Phys. **154**, 064110 (2021).
16. G. Liu, C. Zhang, S. Ciborowski, **A. Asthana**, L. Cheng and K. Bowen, "Mapping the electronic structure of the uranium(VI) dinitride molecule, UN_2", J. Phys. Chem. A **124**, 6486 (2020).
17. **A. Asthana**, J. Liu, and L. Cheng, "Exact two-component equation-of-motion coupled-cluster singles and doubles method using atomic mean-field spin-orbit integrals", J. Chem. Phys. **150**, 074102 (2019).
18. J. Liu, **A. Asthana***, L. Cheng, and D. Mukherjee, "Unitary coupled-cluster based self-consistent polarization propagator theory: A third-order formulation and pilot applications", J. Chem. Phys. **148**, 244110 (2018).
19. J. Liu, Y. Shen, **A. Asthana**, and L. Cheng, "Two-component relativistic coupled-cluster methods using mean-field spin-orbit integrals", J. Chem. Phys. **148**, 034106 (2018).

Presentations

Invited

1. American Chemical Society Spring 2027 Meeting, New Orleans, LA, Mar 2027 (upcoming invited talk; title forthcoming).
2. Midwest Theoretical Chemistry Conference, University of Pittsburgh, Pittsburgh, PA, May 2027 (upcoming invited talk; title forthcoming).

3. (Plenary talk) 65th Sanibel Symposium, Feb 2026, "Quantum algorithms for challenging problems in ground and excited state quantum chemistry", abstract
4. IEEE Quantum Week Sep 2025, "Quantum algorithms for quantum chemistry"
5. The University of Minnesota Apr 2025, "Towards quantum chemistry using quantum computers"
6. The University of Rhode Island Nov 2024, "Towards quantum chemistry using quantum computers" webpage.
7. (Keynote talk) IEEE 2024 Workshop in Chemical Applications of Quantum Computing, Montreal, Canada, "Towards simulating molecular excited states on quantum computers" Sep 2024 webpage.
8. Concordia University Mar 2024, "Quantum chemistry on quantum computers".
9. Questaal Workshop in NREL, Mar 2024, "Quantum chemistry in the era of quantum computing - opportunities and challenges", webpage.
10. American Physical Society National Meeting Minneapolis, MN Mar 2024, "Towards simulating molecular excited states on quantum computers", web.
11. TIFR Mumbai, Dec 2023, "Quantum chemistry in the era of quantum computing - opportunities and challenges".
12. IIT Bombay, Dec 2023, "Quantum chemistry in the era of quantum computing - opportunities and challenges".
13. University of Hyderabad, Nov 2023, "Quantum chemistry in the era of quantum computing - opportunities and challenges".
14. Pacific Northwest National Lab, Nov 2023, "Quantum chemistry in the era of quantum computing, opportunities and challenges".
15. University of North Dakota Physics, Oct 2023,"Towards molecular simulation on quantum computers".
16. (Keynote talk) IEEE Quantum Week, Workshop on Quantum Computing for Natural Sciences: Technology and Applications Sep 2023, "Molecular excited states on quantum computers".
17. University of North Dakota, (Job talk) Apr 2023, "Towards molecular simulation on quantum computers".
18. Quantum Brilliance, Germany, Feb 2023, "Towards molecular simulation on near-term quantum computers".
19. VTQ symposium, Virginia Tech, Feb 2023 "Pulse-level molecular simulation algorithms for near-term quantum computers".
20. VTQ internal seminar, Virginia Tech, Dec 2022, "Path to quantum chemistry on near-term quantum computers".
21. Applied Physics Lab, Johns Hopkins University, Nov 2022, "Pulse-level molecular simulation on transmon qubits".
22. Quantinuum (Honeywell), UK, Oct 2022, "Towards quantum chemistry on near-term quantum computers".
23. Oak Ridge National Lab, Jul 2022, "Pulse-level variational molecular simulation".
24. Virginia Tech, Mar 2021, "Relativistic coupled-cluster methods for heavy-element-containing molecules".
25. IBM Almaden Research Center, Jul 2020, "Relativistic quantum chemistry methods for heavy-element computational chemistry".

Contributed

26. 18th International Congress on Quantum Chemistry (ICQC 2026), Berkeley, CA, Jun 2026, "Quantum Krylov using unitary decomposition for exact and tunable Krylov subspaces".
27. APS March Meeting March 2025, "Advances in molecular excited state methods for quantum computers."
28. Midwest Theoretical Chemistry Conference, Purdue U, IN Jun 2023, "New method for excited-state calculations on near-term quantum computers".
29. American Physical Society National Meeting Las Vegas, NV Mar 2023, "Role of orbital-optimization molecular ground and excited-state calculations on quantum computer (Oral presentation)".
30. American Chemical Society National Meeting Chicago, IL Aug 2022, "Equation-of-motion variational quantum eigensolver method for computing molecular excitation energies, ionization potentials, and electron affinities (Oral presentation)".
31. American Chemical Society National Meeting Chicago, IL Aug 2022, "Minimizing pulse level variational molecular simulation (Oral presentation)".
32. Molecular Quantum Mechanics (MQM), Blacksburg, Jun 2022, "Equation-of-motion variational quantum eigensolver method for computing molecular excitation energies, ionization potentials, and electron affinities (Poster presentation)".

33. Quantum Information Science symposium at Virginia Tech, Apr 2022, "Pulse-level variational molecular simulation (Oral presentation)".
34. American Physical Society National Meeting Chicago, IL Mar 2022, "Minimizing state preparation times in pulse-level variational molecular simulation (Oral presentation)".
35. American Chemical Society National Meeting & Exposition, San Francisco, CA, Aug 2020, "New algorithmic development for relativistic equation-of-motion coupled-cluster method (Poster presentation)".
36. Johns Hopkins University, Dec 2019, "Development of relativistic quantum chemistry methods for molecules containing heavy-elements (Graduate Board Oral talk)".
37. Johns Hopkins University, Mar 2019, "Chemistry at ultracold temperatures (Department seminar)".
38. Indian Institute of Technology Kanpur, Apr 2016, "Extended Wick's theorem, spin-free cumulants, and their role in formulating and analyzing spatial and spin correlation in many-electron systems (master's thesis talk)".

Research Software

QCANT	Open-source Python utilities for near-term quantum computing in chemistry and materials science.
BenchmarkQC	Benchmark Hamiltonians and chemically relevant correlation-regime tests for quantum algorithms.
AutoGen-Wick	Automatic expression generation using Wick's theorem for quantum chemical theories.
CFOUR	Contributor to an international quantum-chemistry software project, including relativistic spin-orbit coupled-cluster methods for heavy-element molecules.

Teaching

Fall 2024, Fall 2026	(Newly developed) Applied Quantum Computing (student rating: 4.7/5)
Spring 2024, Spring 2026	CHEM 471/532, Quantum Mechanics and Its Applications in Chemistry (student rating: 5/5)
Spring 2026	CHEM 221, Fundamentals of Chemistry (co-taught)

Mentoring

Postdoctoral researcher	Dr. Mushir Thodika (since Spring 2026)
Graduate students	Prince Frederick Kwao (since Fall 2023), Sonal Jain (since Spring 2026), Tanya Keshari (since Fall 2026), Hansika Khatri (since Fall 2026)
Undergraduate researcher	Edward Tortorelli (since Summer 2026)
Postdoctoral alumnus	Dr. Srivathsan Poyyapakkam Sundar (2024–2026) Now: postdoctoral researcher at Helmholtz-Zentrum Berlin with Prof. Annika Bande
Undergraduate alumni	Adan Abdihakim (Concordia College), Sean Glaholt (with Prof. Yen Lee Loh)

Services

2027–2029	Elected Chair, American Chemical Society Red River Valley Local Section.
Nov 2024	NSF review panel member (2024)
2024	Guest editor in Quantum Science and Technology, IOP focus issue on quantum computing applications in chemistry.
2023	Member of the editorial board for the Quantum Computing paper track for ISC High Performance 2024.
Jan-May 2023	Co-organized Virginia Tech Quantum (VTQ) seminar series
2020–present	Reviewer for npj Quantum Information (Nature), Chemical Science (RSC), Journal of Chemical Theory and Computation (ACS), ACS Central Science, and Journal of Chemical Physics (AIP).
April 2022	Co-organized Virginia Tech Quantum (VTQ) symposium, a two-day symposium with more than 40 participants from the Department of Physics, Computer Science and Chemistry

Achievements

2026	DOE Early Career Research Program Award.
2026	American Chemical Society Petroleum Research Fund (ACS PRF) Doctoral New Investigator Award.
Dec 2024	UND Early Career Scholar Award.
Nov 2024	NSF EPSCoR Research Fellow Award.
May 2016	Best master's project award, Department of Chemistry, IIT Kanpur.
2011-2016	Awarded *INSPIRE-SHE Scholarship* worth $6000, Department of Science and Technology, Government of India.

Media

- "Computing the future at UND", *University Letter* (UND), Aug. 18, 2026
- "UND professors receive NSF funding for national security research", *UND Today*, Jan. 16, 2025